



Semester: Jan 2025 – April 2025		
Maximum Marks: 50	Examination: End-Semester Examination	Duration: 2 Hrs.
Programme code: 003 Programme: Btech Honours in Data Science for Emerging Analytics	Class: S.Y. Btech	Semester: IV (SVU 2023)
Institute/School/Department: K. J. Somaiya School of Engineering	Name of the department: EXTC	
Course Code: 216h27C401	Name of the Course: Advanced Data Analytics (Honours Programme in Data Science for Emerging Analytics)	
Instructions: 1) Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question Statement	Max. Marks																																								
Q.1	Attempt any one																																									
i)	Consider a simple time series dataset where we observe daily stock prices for 10 days: $X = \{10, 12, 19, 18, 11, 13, 14, 15, 16, 18\}$. Calculate the autocorrelation at lag 1 and lag 2	10																																								
ii)	What are the terms involved in decomposition of time series. Explain the characteristics of scenario when will you use additive and multiplicative models for data? Write the mathematical equations for same.	10																																								
Q.2	Attempt any one	10																																								
i)	The stem and leaf plot below shows the grade point averages of 18 students. The digit in the stem represents the ones and the digit in the leaf represents the tenths. So for example $0 8 = 0.8$, $1 2 = 1.2$ and so on. <table><tr><td>0</td><td>8</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>1</td><td>2</td><td>4</td><td>4</td><td>6</td><td>7</td><td></td><td></td></tr><tr><td>2</td><td>0</td><td>4</td><td>5</td><td>5</td><td>6</td><td>7</td><td>9</td></tr><tr><td>3</td><td>2</td><td>3</td><td>5</td><td>8</td><td></td><td></td><td></td></tr><tr><td>4</td><td>0</td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table> a) What is the range of the data in the stem and leaf plot? b) How many students have a grade of 2 or more? c) What is the mode of the grades? d) What is the median of the grades? e) What is the sample mean of the grades?	0	8							1	2	4	4	6	7			2	0	4	5	5	6	7	9	3	2	3	5	8				4	0							
0	8																																									
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2	0	4	5	5	6	7	9																																			
3	2	3	5	8																																						
4	0																																									
ii)	Following is the data of number of cars on road daily and number of accidents, monthly.																																									



		<table><tr><th>Number of cars(daily)</th><th>Number of accidents(monthly)</th></tr><tr><td>2000</td><td>15</td></tr><tr><td>2300</td><td>27</td></tr><tr><td>2500</td><td>20</td></tr><tr><td>2600</td><td>21</td></tr><tr><td>2800</td><td>31</td></tr><tr><td>3000</td><td>16</td></tr></table>	Number of cars(daily)	Number of accidents(monthly)	2000	15	2300	27	2500	20	2600	21	2800	31	3000	16	
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2000	15																
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a) Sketch a scatterplot. b) Compute the coefficients of the linear regression line, $y = b_1x + b_0$. Estimate the number of accidents in a month when the number of cars using the highway is 2400.																	
Q.3 Attempt any one			10														
i) Explain the steps in applying the SARIMA model to a data set.																	
ii) Apply Single Exponential smoothing for the following data:																	
<table><tr><th>Month</th><th>Sale Quantity</th></tr><tr><td>1</td><td>30</td></tr><tr><td>2</td><td>44</td></tr><tr><td>3</td><td>32</td></tr><tr><td>4</td><td>42</td></tr><tr><td>5</td><td>30</td></tr><tr><td>6</td><td>43</td></tr></table>			Month	Sale Quantity	1	30	2	44	3	32	4	42	5	30	6	43	
Month	Sale Quantity																
1	30																
2	44																
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5	30																
6	43																
Q.4 Attempt the following			10														
What is the difference between data and information? What are the differences between spatial and attribute data with respect to Map Anatomy?																	
Q.5 Attempt the following			10														
What is Power BI? List the different features of Power BI and explain in detail.																	